Bachelor of Computer Applications (Hons. Game Development)
Semester- III

Subject: Database Management System

Assignment 5: Database Creation and Table Implementation

By
Name: FIDHA
Reg. No.: 2511021240021

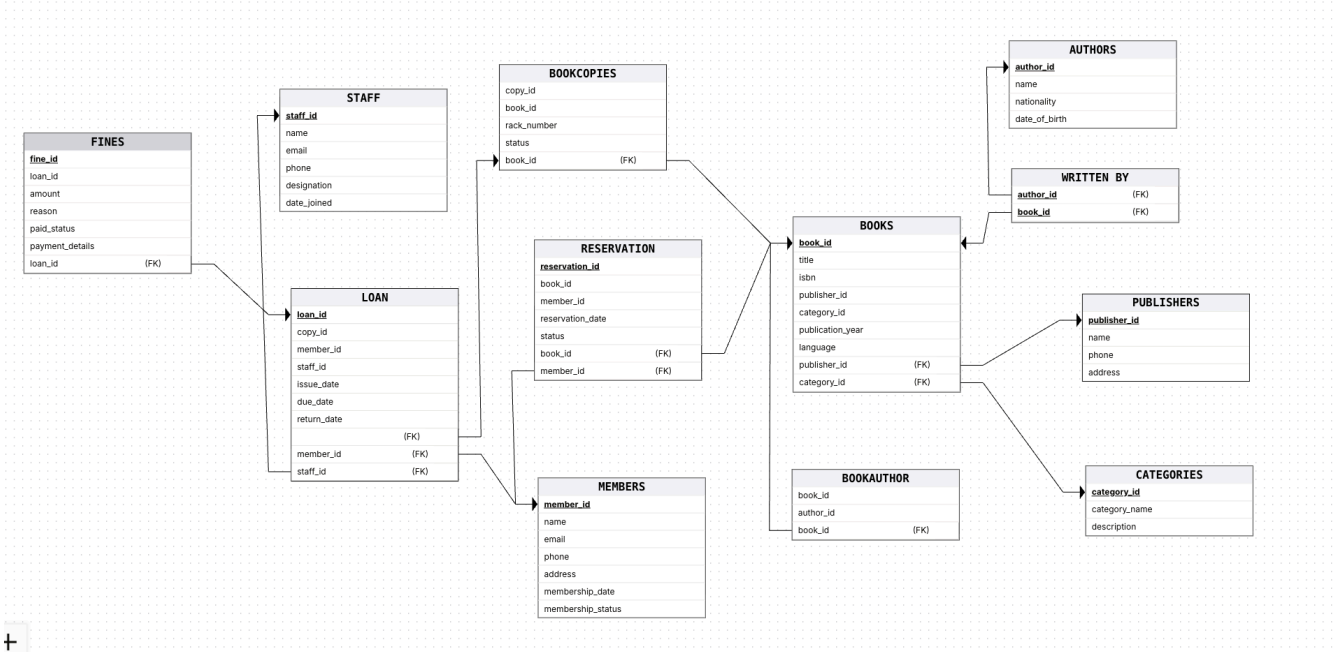
Faculty Name: Renita Joy
Faculty Signature:_____

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru - 562 106
October 2026

1. Overview

The DDL below creates the database and the 11 tables from Assignments 1 and 2. The statements use MySQL 8.0 syntax. Tables are created in dependency order so referenced tables exist before foreign keys are added.

2. Relationship diagram



3. SQL DDL commands

```
CREATE DATABASE IF NOT EXISTS LibraryManagement;

USE LibraryManagement;

CREATE TABLE Members (
 member_id INT PRIMARY KEY,
 name VARCHAR(100) NOT NULL,
 email VARCHAR(150) NOT NULL UNIQUE,
 phone VARCHAR(20) UNIQUE,
 address VARCHAR(255),
 membership_date DATE NOT NULL DEFAULT (CURRENT_DATE),
 membership_status VARCHAR(20) NOT NULL DEFAULT 'Active',
 CONSTRAINT chk_member_status CHECK (membership_status IN ('Active','Suspended','Expired')) );
```

```
CREATE TABLE Staff (
 staff_id INT PRIMARY KEY,
 name VARCHAR(100) NOT NULL,
 email VARCHAR(150) NOT NULL UNIQUE,
 phone VARCHAR(20) UNIQUE,
 designation VARCHAR(80) NOT NULL,
 date_joined DATE NOT NULL
);


CREATE TABLE Publishers (
 publisher_id INT PRIMARY KEY,
 name VARCHAR(150) NOT NULL UNIQUE,
 phone VARCHAR(20),
 address VARCHAR(255)
);


CREATE TABLE Authors (
 author_id INT PRIMARY KEY,
 name VARCHAR(120) NOT NULL,
 nationality VARCHAR(80),
 date_of_birth DATE
);


CREATE TABLE Categories (
 category_id INT PRIMARY KEY,
 category_name VARCHAR(80) NOT NULL UNIQUE,
 description VARCHAR(255)
);


CREATE TABLE Books (
 book_id INT PRIMARY KEY,
 title VARCHAR(200) NOT NULL,
 isbn VARCHAR(20) NOT NULL UNIQUE,
 publisher_id INT NOT NULL,
 category_id INT NOT NULL,
 publication_year INT,
 language VARCHAR(50) NOT NULL DEFAULT 'English',
 CONSTRAINT fk_books_publisher FOREIGN KEY (publisher_id) REFERENCES Publishers(publisher_id),
CONSTRAINT fk_books_category FOREIGN KEY (category_id) REFERENCES Categories(category_id),
CONSTRAINT chk_publication_year CHECK (publication_year IS NULL OR publication_year BETWEEN 1450
AND 2100) );


CREATE TABLE BookCopies (
```

```
 copy_id INT PRIMARY KEY,
 book_id INT NOT NULL,
 rack_number VARCHAR(20) NOT NULL,
 status VARCHAR(20) NOT NULL DEFAULT 'Available',
 CONSTRAINT fk_copies_book FOREIGN KEY (book_id) REFERENCES Books(book_id),
 CONSTRAINT chk_copy_status CHECK (status IN ('Available','Issued','Damaged','Lost'))
);


CREATE TABLE Book_Author (
 book_id INT NOT NULL,
 author_id INT NOT NULL,
 PRIMARY KEY (book_id, author_id),
 CONSTRAINT fk_ba_book FOREIGN KEY (book_id) REFERENCES Books(book_id),
 CONSTRAINT fk_ba_author FOREIGN KEY (author_id) REFERENCES Authors(author_id)
);

CREATE TABLE Loans (
 loan_id INT PRIMARY KEY,
 copy_id INT NOT NULL,
 member_id INT NOT NULL,
 staff_id INT NOT NULL,
 issue_date DATE NOT NULL,
 due_date DATE NOT NULL,
 return_date DATE NULL,
 CONSTRAINT fk_loans_copy FOREIGN KEY (copy_id) REFERENCES BookCopies(copy_id),
 CONSTRAINT fk_loans_member FOREIGN KEY (member_id) REFERENCES Members(member_id),
 CONSTRAINT fk_loans_staff FOREIGN KEY (staff_id) REFERENCES Staff(staff_id),
 CONSTRAINT chk_loan_dates CHECK (due_date &gt;= issue_date AND (return_date IS NULL OR return_date
&gt;= issue_date)) );


CREATE TABLE Reservations (
 reservation_id INT PRIMARY KEY,
 book_id INT NOT NULL,
 member_id INT NOT NULL,
 reservation_date DATE NOT NULL DEFAULT (CURRENT_DATE),
 status VARCHAR(20) NOT NULL DEFAULT 'Pending',
 CONSTRAINT fk_res_book FOREIGN KEY (book_id) REFERENCES Books(book_id),
 CONSTRAINT fk_res_member FOREIGN KEY (member_id) REFERENCES Members(member_id),
 CONSTRAINT chk_res_status CHECK (status IN ('Pending','Fulfilled','Cancelled'))
);


CREATE TABLE Fines (
 fine_id INT PRIMARY KEY,
 loan_id INT NOT NULL UNIQUE,
 amount DECIMAL(8,2) NOT NULL DEFAULT 0.00,
 reason VARCHAR(20) NOT NULL,
```

```
 paid_status VARCHAR(20) NOT NULL DEFAULT 'Unpaid',
 payment_details DATE NULL,
 CONSTRAINT fk_fines_loan FOREIGN KEY (loan_id) REFERENCES Loans(loan_id),
 CONSTRAINT chk_fine_amount CHECK (amount &gt;= 0),
 CONSTRAINT chk_fine_reason CHECK (reason IN ('Late return','Damage','Lost')),
 CONSTRAINT chk_fine_paid CHECK (paid_status IN ('Paid','Unpaid')),
 CONSTRAINT chk_paid_date CHECK ((paid_status = 'Unpaid' AND payment_details IS NULL) OR paid_status
= 'Paid') );
```

4. Implementation notes

- Members, Staff, Publishers, Authors, and Categories are master tables.
- Books reference Publishers and Categories.
- BookCopies references Books.
- Book_Author uses a composite primary key and two foreign keys to resolve the many-to-many relationship.
- Loans references BookCopies, Members, and Staff.
- Reservations references Books and Members.
- Fines references Loans, with UNIQUE(loan_id) enforcing the one-to-one rule from the original design.